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A Microscopic Traffic Simulation Study of Autonomous Vehicle Penetration Rate Effects on Traffic Flow Across Different Control Models and Scenarios

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Abstract

Autonomous vehicle (AV) development is a rapidly progressing field. Understanding how AVs will reshape traffic flow before they are deployed at scale is essential for guiding their introduction. Simulation makes this possible by letting different AV control models be evaluated side by side under controlled conditions, exploring possible futures that cannot yet be observed on real roads. The goal of this thesis was to evaluate how increasing levels of AV penetration ranging from 0% to 100% influence the efficiency and stability of various road networks.

To accomplish this, the Simulation of Urban Mobility (SUMO) platform was used to implement and test different AV control models. Five control models were evaluated, a parameter-tuned IDM, a Krauß-based model, ACC, CACC and GLOSA, across four simulated environments, a closed ring road, an open multi-lane freeway based on the real-world Interstate 80, a real-world urban network of Bologna and a custom intersection grid. These models were deployed across these environments to analyze how AV penetration affects traffic flow across several metrics, including average speed and vehicle throughput. The results demonstrate that AVs generally improve the efficiency and stability of traffic flow, with the size of the effect depending strongly on both the control model used and the complexity of the road layout. Higher penetration rates dissipate the stop-and-go waves that form in dense traffic, as shown in the ring scenario.

Sammanfattning

Utvecklingen av autonoma bilar (AV) är ett snabbt framskridande område. Att förstå hur autonoma fordon kommer att förändra trafikflödet innan de tas i bruk i stor skala är avgörande för att vägleda deras införande. Simulering möjliggör detta genom att olika styrmodeller för autonoma fordon kan utvärderas sida vid sida under kontrollerade förhållanden, vilket låter oss utforska tänkbara framtider som ännu inte kan observeras på verkliga vägar. Målet med denna uppsats var att utvärdera hur ökande nivåer av AV-genomträngning från 0% till 100% påverkar effektiviteten och stabiliteten hos olika vägtyper.

För att uppnå detta användes plattformen Simulation of Urban Mobility (SUMO) för att implementera och testa olika styrmodeller för autonoma fordon. Fem styrmodeller utvärderades, en parameterjusterad IDM, en Krauß-baserad modell, ACC, CACC och GLOSA, i fyra simulerade miljöer, en sluten ringväg, en öppen flerfilig motorväg baserad på verkliga Interstate 80, ett verkligt stadsnät från Bologna och ett specialbyggt korsningsnät. Dessa modeller tillämpades i dessa miljöer för att analysera hur AV-genomträngning påverkar trafikflödet över flera mått, däribland genomsnittshastighet och fordonsflöde. Resultaten visar att autonoma fordon generellt förbättrar trafikflödets effektivitet och stabilitet, där effektens storlek beror starkt på både den använda styrmodellen och vägnätets komplexitet. Högre genomträngningsgrader dämpar de start- och stoppvågor som uppstår i tät trafik, vilket visas i ringscenariot.

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Chapter 1

Introduction

In recent years autonomous vehicle (AV) development has progressed at a rapid pace with the ultimate goal of real-world usage. As of now, multiple cities have fully autonomous vehicles deployed and many more areas have some level of autonomy allowed in vehicles [1]. Autonomy in a vehicle does not necessarily mean completely self-driving but instead the autonomy is usually classified on a scale from 0-5 with 0 being no autonomy and 1 to 5 ranging from basic cruise control to full self-driving autonomy[2].

The current focus of AV development is mainly to increase safety and create a more convenient driving experience[3] while they might also optimize traffic flow when implemented on a larger scale. The impact of the AVs on traffic flow might yet be difficult to study in real-world scenarios though the results of such a study would provide critical information about future deployment. The study of simulating AVs in traffic has already emerged though a consensus on their theoretical impact is not found and their impact is still uncertain. A central part of the impact of AVs is their penetration rate (percentage of AVs), which is commonly used in studies such as Stern et al.[4]. It gives insight to the performance of AVs and is relevant since real-world deployment would result in a gradual rise in the penetration rate but a lower rate might not grant any advantages, as seen in certain metrics in Lu et al.[5]. Therefore, it is interesting to, for example, find a penetration rate where the impact is at the critical point where their impact per car is the most significant.

However as of right now, previous studies have used a wide range of different models for AVs but the result of each is inevitably tied to the scenarios and simulation methods. These different models provide an interesting point to study since it might give us more accurate insight as to how the real-world deployment of AVs compare to each other and how they will impact traffic.

1.1 Research Question

This thesis sets out to study the impact of the different previous implementations of AV models to accurately simulate different traffic scenarios and larger traffic contexts.

Our research question is divided into three sub-questions which aim to answer our core research question which is as follows: What does microscopic traffic simulation reveal about the potential of autonomous vehicles to improve traffic flow?

The three resulting sub-questions that aim to answer our core research question are:

1. How do different AV control models (parameter-tuned car-following, ACC, CACC, GLOSA) compare in their effect on traffic flow?

2. How do different levels of AV penetration (0%, 5%, 10%, ..., 100%) affect traffic flow?

3. How does the road network type (closed loop, free-flow highway, signalised urban, and intersection grid) influence the effect of AV adoption on traffic flow?

1.2 Hypothesis

Our hypothesis is that an increased AV penetration will generally result in increased traffic flow, although the magnitude of the effect is expected to vary between AV control models. This is based upon our initial beliefs that the reduced reaction times, consistent behavior and use of optimal driving strategies will result in an improved flow of traffic. Our assumptions are based on previous studies such as Lu et al.[5] where an increased penetration rate leads to an increase in maximum flow among other metrics. The impact will likely reach maximum efficiency with high penetration rates (>90%) but the largest improvements will be seen from 50% to 80% when the AVs control most of the traffic which provides stability to the system as seen in Katsaros et al.[6]. The exact size of the improvement is expected to depend on the specific AV control model used.

1.3 Scope and Delimitations

The scope of this thesis is to evaluate the effect of increasing AV penetration on traffic flow across a set of road networks using microscopic traffic simulation. The AV models studied are simplified representations of fully autonomous vehicles, where each model represents a distinct driving strategy that attempts to mimic a real-world implementation. Safety functionality is assumed to be satisfied and is not evaluated. Only passenger vehicles are considered, and each AV model is tested independently such that no simulation contains a mixture of different AV types.

1.4 AI Usage

Generative AI was used in the writing process of this thesis for language editing and proofreading purposes. It was also used during development of the simulation code as a tool for debugging and syntax guidance. All AI output was verified before being used.

1.5 Outline

This thesis is structured as follows; Firstly, *Chapter 2* provides necessary background information on which the thesis is based upon. The focus is especially on AV simulation and how the simulation program SUMO functions is necessary to understand the theory of the thesis. The literature review that ends this chapter provides the previous studies on the subject which models which are used in the simulations.

Chapter 3 describes how the intent of this thesis is accomplished. It is separated into three parts, two of which cover different aspects of the approach and the last which describes how the results were gathered.

Chapter 4 presents the results gathered. The chapter is separated into 4 sections each covering a different scenario making it possible to compare the different models.

Chapter 5 discusses the findings from the previous chapter and is divided into 5 different sections, three covering our sub-questions to our research question, one answering the core research questions and the final on discussing limitations.

Chapter 6 concludes the thesis and the important findings are presented together with one section for possible future work connected to the thesis.

Chapter 2

Background

The two primary areas this thesis intersects are autonomous vehicles and traffic simulation, where traffic simulation could be considered a subfield of the broader field of crowd simulation. AVs create a subject to study while traffic simulation provides the framework needed to model and evaluate their behavior and impact.

2.1 Autonomous Vehicles

Autonomous vehicles (AVs) are often classified into different levels of automation ranging from 0-5 with 0 being no automation and 5 being fully autonomous. For this thesis Adaptive Cruise Control (ACC), Cooperative Adaptive Cruise Control (CACC) and green light optimal speed advisory (GLOSA) will be studied, all of these create a level 1 autonomy in the vehicle since they each control one feature (accelerating/decelerating). Level 5 automation will also be studied with two different models that try to mimic how a fully autonomous vehicle would act. When referencing autonomous vehicles all the levels above 0 are assumed[2].

Current real-world implementations of fully autonomous vehicles (level 5) are not yet capable of effectively improving traffic but instead aim to increase safety so that a more widespread introduction can be achieved[3]. This thesis assumes the safety functionalities of the AVs are satisfied and focuses instead on evaluating the decision models that determine AVs ability to improve traffic flow.

2.2 Traffic Theory

2.2.1 Wave formation

The theory behind the formation of waves has been studied for more than 70 years and there are a few theories which summarize the phenomenon. The resulting waves can be categorized into two sorts, waves of stopping and waves of starting. The first being a result of a traffic situation where drivers are forced to decelerate which creates a wave downstream of cars decelerating. The other being when a temporary hindrance is cleared and cars start to accelerate which also sends a wave downstream of cars accelerating. [7]

When cars are in a state of free flow traffic there are two effects working upon the flow, there is a stabilizing effect which is caused by driving preventively but also a destabilizing which is caused by drivers slowing down due to increasing traffic density. When the local traffic density of a highway reaches a certain threshold spontaneous waves are created by individual drivers decelerating which cause what is called phantom traffic jams. [8]

Autonomous vehicles reduce the formation of waves since their reaction is instantaneous unlike that of a human driver while also acting with less impulsiveness leading to a more stable flow. This can be observed in Lu et al. [5] where the flow and speed of AVs is less affected by higher vehicle densities than human drivers.

A related concept is string stability, which describes whether a disturbance introduced by one vehicle is damped or amplified as it propagates upstream through a line of following vehicles [9, 10]. A string-stable stream absorbs such disturbances, whereas a string-unstable one lets them grow into stop-and-go waves.

2.3 Measurements

Measuring traffic flow is done by a number of different metrics each granting insight on different aspects which impact traffic flow and how the network is performing. To calculate these metrics, this study adopts the methodology outlined by Lu et al.[5].

Vehicle speed measures the velocity of each vehicle in meters per second (m/s). It reflects the freedom of the traffic flow which allows for higher vehicle speeds. The average vehicle speed V is calculated using the following formula,

where l is the driven distance and t is the driven time for each vehicle i:

$$V = \frac{\sum_{i=1}^{n} l_i}{\sum_{i=1}^{n} t_i}$$

Vehicle throughput reflects how effectively the measured traffic scenario is performing and gives a good overall metric. It is measured in vehicles per hour (veh/h) and is calculated by multiplying vehicle density ρ and vehicle speed V for each section i of road:

$$Q = \sum_{i=1}^{n} \rho_i V_i$$

2.4 Simulating in SUMO

Simulation of Urban Mobility (SUMO) is an open source microscopic traffic simulation package developed primarily by the German Aerospace Center (DLR) [11]. SUMO models each vehicle individually as it moves through a road network, computing its position, speed, and lane at every simulation step. A simulation requires two primary inputs: a road network, which defines the topology of roads, junctions, and traffic signals, and a traffic demand, which specifies which vehicles enter the network and what route they follow.

The microscopic simulation makes it well suited for simulating the difference in performance between AVs and human drivers, since changes to individual vehicle behavior are directly manifested in the overall performance of the traffic network. SUMO is commonly used in traffic simulating research such as Porfyri et al.[12].

2.4.1 Car-Following Models

The longitudinal movement of each vehicle in SUMO is governed by a car-following model, which determines the speed a vehicle adopts based on its own state and the state of the vehicle directly ahead. SUMO ships with several such models; the two most relevant to this thesis are the Intelligent Driver Model (IDM) and the Krauß model.

The Intelligent Driver Model of Treiber, Hennecke, and Helbing [13] defines a desired dynamic gap that grows smoothly with speed and with the relative speed to the leader, and derives the acceleration from the difference between the current state and this desired state. The resulting acceleration profiles are continuous and physically plausible, and the model has been calibrated

against empirical trajectory data in a number of studies. IDM does not include an explicit driver imperfection term; variability between drivers is instead introduced by varying parameters such as the desired speed, the time headway, or the comfortable deceleration.

SUMO's default model is that of Krauß [14], which computes a safe speed guaranteeing that the follower can stop before colliding with its leader even if the leader brakes at maximum deceleration, and then draws the actual speed from an interval below this safe speed. The width of that interval is controlled by the driver imperfection parameter `sigma`, which introduces stochastic variation and represents human reaction noise. The remaining parameters are the maximum acceleration, the maximum deceleration, the desired minimum time headway (`tau`), and the minimum gap in a standing queue (`minGap`).

In addition to these general-purpose models, SUMO provides an Adaptive Cruise Control (ACC) model and a Cooperative Adaptive Cruise Control (CACC) model. Unlike IDM and Krauß, these are not derived from analytical principles but calibrated directly against field measurements of production ACC hardware [9], and the CACC variant additionally adapts its behaviour based on whether the preceding vehicle is also CACC-equipped. The car-following model is specified per vehicle type, so models of different families can coexist in the same simulation.

2.4.2 Lane-Changing Model

Lateral movement is handled by a separate lane-changing model. The default model, LC2013, evaluates lane changes based on three motivations: strategic changes to reach the correct lane for an upcoming turn, cooperative changes to make room for other vehicles, and speed-gain changes to move into a faster lane [15].

2.4.3 TraCI

SUMO provides an interface called TraCI (Traffic Control Interface) that allows an external program to read and modify the state of a running simulation at each time step [11]. This includes reading vehicle positions and speeds, overriding vehicle speeds, and changing traffic signal states. TraCI can also be used to implement custom vehicle controllers by externally setting vehicle behaviour during the simulation. In this thesis, however, TraCI is used only in the ring-road scenario to introduce a controlled speed perturbation after the warm-up period, the AV behaviours themselves are defined through SUMO

vehicle types, built-in car-following models, and the built-in GLOSA device.

There have been several approaches to simulating autonomous vehicles in microscopic traffic simulators such as SUMO. These approaches differ primarily in how they represent the driving behavior of the autonomous vehicle. This section presents the studies most relevant to our work.

2.4.4 Parameter-Based Models

A common method for representing AVs in SUMO is to modify the parameters of an existing car-following model. Lu et al. [5] investigated the impact of AVs on urban road capacity using SUMO's default Krauß model. In their study, the driver imperfection parameter was set to zero for AVs, and the desired time headway was reduced from its default value. Six penetration rates ranging from 0% to 100% were evaluated on a grid network and a real-world network of Budapest. The results indicated that road capacity increased with AV penetration. This approach does not model any specific AV capability such as communication or coordination; it represents only the expectation that automated vehicles will exhibit less randomness and maintain shorter headways than human drivers.

2.4.5 Cooperative Adaptive Cruise Control

Cooperative Adaptive Cruise Control (CACC) is a car-following model that represents vehicles which share acceleration data with nearby vehicles through vehicle-to-vehicle communication. Porfyri, Mintsis, and Mitsakis [12] integrated ACC and CACC car-following models into SUMO, building on the empirical controller developed by Milanés and Shladover [9] from experiments conducted at the California PATH programme. A notable property of the CACC model is that it adapts its following behaviour based on the type of the preceding vehicle. When the preceding vehicle is also CACC-equipped, a shorter time gap is used; when the preceding vehicle is human-driven, the model falls back to a more conservative gap. Porfyri, Mintsis, and Mitsakis [12] evaluated these models on a single-lane freeway stretch and a ring road, finding that ACC and CACC vehicles improved traffic flow characteristics compared to manual driving. In particular, ACC vehicles were shown to enhance the stability of traffic flow and eliminate stop-and-go waves on the ring road. Their results also indicated that the effect increased with penetration rate, as a higher proportion of equipped vehicle pairs emerged.

2.4.6 Green Light Optimal Speed Advisory

Green Light Optimal Speed Advisory (GLOSA) is a vehicle-to-infrastructure application in which vehicles receive signal phase and timing information from upcoming traffic lights and adjust their speed accordingly. The objective is to arrive at the intersection at the moment the signal turns green, thereby avoiding unnecessary stops. Katsaros et al. [6] implemented and evaluated a GLOSA system using SUMO as the traffic simulator, coupled with a network communication simulator to model V2I communication. Their results showed reductions in both fuel consumption and stop time at signalised intersections, with the magnitude of the effect depending on the penetration rate of equipped vehicles.

2.4.7 Traffic Flow Stabilisation Through Coordinated Control

An alternative approach to AV simulation involves overriding SUMO's built-in car-following model entirely and controlling individual vehicles through the TraCI interface. Wu et al. [10] developed Flow, a computational framework for mixed-autonomy traffic research built on SUMO. Using reinforcement learning, they trained AV controllers that determined the speed of each controlled vehicle at every simulation step, based on observations of the surrounding traffic. In a ring road experiment with 22 vehicles, replacing a single human-driven vehicle with a trained AV eliminated the stop-and-go waves that otherwise emerged among the remaining human drivers, substantially increasing the average speed of all vehicles. In a complementary physical field experiment, Stern et al. [4] placed a single controlled vehicle among 21 human drivers on a circular track. The controlled vehicle was able to dampen stop-and-go waves and reduce overall fuel consumption by up to 40%.

2.4.8 Autonomous Intersection Management

Dresner and Stone [16] proposed a reservation-based protocol for managing intersections without traffic signals, referred to as Autonomous Intersection Management (AIM). In their approach, vehicles approaching an intersection communicate with a central intersection manager to request a time-space slot for crossing. The manager assigns slots such that no two vehicles occupy the same space at the same time. Simulations show substantial throughput improvements compared to conventional traffic signals, but the protocol required a high proportion of equipped vehicles to function effectively. In mixed traffic

scenarios where human-driven vehicles are present, the system must account for vehicles that do not participate in the reservation protocol, which limits the achievable improvement at lower penetration rates.

Chapter 3

Method

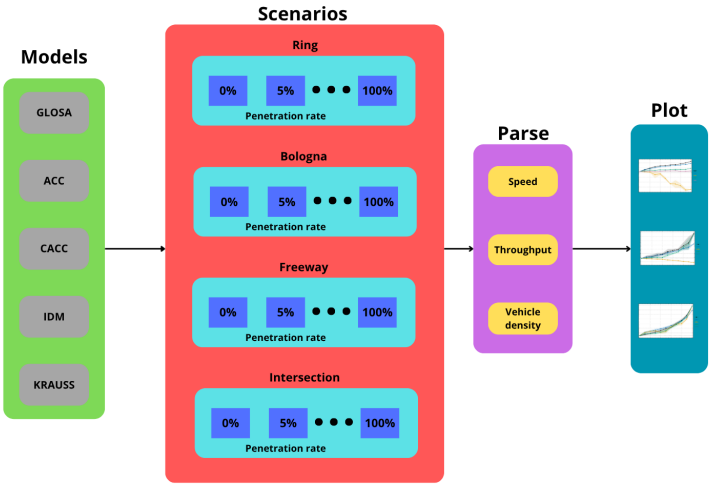


Figure 3.1: This chart represents the methodology of this thesis. The models used are inserted to each scenario where they are run at penetration rates ranging from 0% to 100% with 5% intervals. Afterwards the metrics are parsed through the data from the simulation and thereafter plotted.

3.1 AV Models

Each AV model is implemented as a vehicle type in SUMO and assigned to vehicles at departure time based on the penetration rate of the given experiment run. The penetration rate determines the probability that a departing vehicle is assigned an AV type rather than the human type. The same random seed is

used across models for a given penetration rate to ensure that the set of vehicles designated as AVs is identical across tiers. The IDM-based AV and Krauß tiers share the same shortened-headway AV parameter envelope, while ACC and CACC use the acceleration and maximum-speed parameters from Porfyri, Mintsis, and Mitsakis [12]. GLOSA is implemented as a speed-advisory device layered on top of the human-parameter IDM model.

3.1.1 Human Baseline

The human driver model follows the Intelligent Driver Model (IDM) of Treiber, Hennecke, and Helbing [13] rather than the stochastic Krauß model used as SUMO's default. IDM produces smoother, physically plausible acceleration profiles and has been widely validated against empirical trajectory data, making it a more realistic reference point against which to measure AV-induced improvements. Human drivers are given a desired time headway of $\texttt{tau} = 1.0\ \mathrm{s}$, a minimum standing gap of $\texttt{minGap} = 2.5\ \mathrm{m}$, and a speed deviation of $\texttt{speedDev} = 0.1$, which introduces a 10% Gaussian spread around the desired speed to represent heterogeneity in driver aggression and desired cruising speed. Maximum acceleration and deceleration are set to $2.6\ \mathrm{m/s^2}$ and $2.0\ \mathrm{m/s^2}$ respectively, reflecting comfortable (rather than emergency) driving dynamics. The vehicle type definition is:

```
<vType id="human" carFollowModel="IDM"
       tau="1.0" minGap="2.5" speedDev="0.1"
       accel="2.6" decel="2.0" length="5"/>
```

Throughout this thesis "human baseline" refers to a simulation in which every vehicle is of this type, i.e. the 0% penetration point of any AV model sweep.

3.1.2 IDM-Based AV (Basic)

The basic AV model represents a realistic deterministic AV: one that behaves as a human-style IDM driver but without reaction noise and with the tighter headways and faster reflexes afforded by automation. It shares its car-following model with the human baseline, which isolates the effect of the AV-specific parameter choices from the effect of switching car-following models. The desired time headway is reduced from $1.0\ \mathrm{s}$ to $\texttt{tau} = 0.6\ \mathrm{s}$, the minimum standing gap from $2.5\ \mathrm{m}$ to $\texttt{minGap} = 1.5\ \mathrm{m}$, and the speed deviation is set to zero so that all AVs of this type drive at exactly the desired speed. IDM has no

explicit driver imperfection parameter, so the only stochastic element present in the human baseline, the `speedDev` spread, is removed.

```
<vType id="av_basic" carFollowModel="IDM"
       tau="0.6" minGap="1.5" speedDev="0.0"
       accel="3.5" decel="2.2" length="5"/>
```

No external control code is required for this tier. The behaviour is determined entirely by the parameter values and SUMO's internal IDM logic. This model serves as the primary reference AV in the sweep, and all remaining tiers are interpreted as additive refinements to it.

3.1.3 Krauß AV (Comparison Tier)

To relate our results to the large body of prior work that represents AVs by zeroing the driver imperfection parameter of SUMO's default Krauß model, most prominently Lu et al. [5], a Krauß-based AV tier was included as a direct comparison. The parameters mirror those used by Lu et al.: driver imperfection `sigma` $= 0$, desired time headway `tau` $= 0.6$ s, and minimum standing gap `minGap` $= 1.5$ m. Unlike IDM, the Krauß model admits an explicit stochastic term; setting it to zero yields fully deterministic car-following.

```
<vType id="av_krauss" carFollowModel="Krauss
    "
       sigma="0.0" tau="0.6" minGap="1.5"
          speedDev="0.0"
       accel="3.5" decel="2.2" length="5"/>
```

This tier is not presented as a candidate model of a real AV but rather as a literature-standard idealisation whose results can be compared directly against earlier studies and against the IDM-based basic tier.

3.1.4 ACC Model

The ACC model uses SUMO's built-in Adaptive Cruise Control car-following model. The implementation in SUMO is due to Porfyri, Mintsis, and Mitsakis [12], who integrated the empirical controller of Milanés and Shladover [9] calibrated against field measurements of production ACC hardware at the California PATH programme. The parameter values used here follow the ring-road setup of Porfyri, Mintsis, and Mitsakis [12]: a bounded maximum acceleration of `accel` $= 1.0$ m/s^2, a maximum speed of `maxSpeed` $= 33.33$ m/s (120 km/h), a desired time gap of `tau` $= 1.5$ s, and a minimum standing gap of

minGap $= 2.0$ m. Unlike the IDM-based tiers, the deceleration is not overridden; the controller's empirically calibrated internal response is used directly.

```
<vType id="av_acc" carFollowModel="ACC"
       accel="1.0" maxSpeed="33.33"
       tau="1.5" minGap="2.0"
       speedDev="0.0" length="5"/>
```

These parameter values differ noticeably from those used by the human baseline and the IDM-based AV tiers, in particular the considerably lower bounded acceleration. Direct numerical comparisons between the ACC tier and the IDM-based tiers should therefore be treated with caution, since differences in their longitudinal envelope will contribute to differences in measured outcomes. The settings are nevertheless kept as in Porfyri, Mintsis, and Mitsakis [12] so that the ACC tier remains a useful reference point for the wave-stabilisation results reported in that study and in the broader ACC literature.

3.1.5 CACC Model

The CACC model uses SUMO's built-in Cooperative Adaptive Cruise Control car-following model, also from Milanés and Shladover [9] and integrated into SUMO by Porfyri, Mintsis, and Mitsakis [12]. The longitudinal dynamics envelope matches the ACC tier (accel $= 1.0$ m/s^2, maxSpeed $= 33.33$ m/s), reflecting the convention of Porfyri, Mintsis, and Mitsakis [12] that ACC and CACC vehicles share the same bounded acceleration and maximum speed so that any difference between the two tiers is attributable to the cooperative component rather than to the vehicle dynamics.

The CACC model determines its following behaviour based on the type of the preceding vehicle. When the predecessor is also of type CACC, the model uses a communication-enabled controller with a shorter time gap of tau $= 0.6$ s and a minimum standing gap of minGap $= 1.0$ m. When the predecessor is not CACC-equipped, the model falls back to a standard ACC controller with a longer gap, set here via headwayTimeACC $= 1.5$ s to match the ACC tier. This switching is handled internally by SUMO and requires no external code. As a consequence, the effective following distance of CACC vehicles decreases as the penetration rate increases and more CACC-to-CACC pairs form in the traffic stream.

```
<vType id="av_cacc" carFollowModel="CACC"
       accel="1.0" maxSpeed="33.33"
       tau="0.6" minGap="1.0"
```

```
        speedDev="0.0" length="5">
  <param key="headwayTimeACC" value="1.5"/>
</vType>
```

As with the ACC tier, the CACC envelope follows Porfyri, Mintsis, and Mitsakis [12] rather than the parameters used by the human baseline and the IDM-based AV tiers. Cross-tier comparisons should therefore be read with the same caveat: differences in measured outcomes between CACC and the IDM-based tiers reflect both the cooperative behaviour and the differing acceleration envelope. The chosen settings make the CACC results directly comparable to those of Porfyri, Mintsis, and Mitsakis [12] and the broader CACC literature, which is the primary reason for adopting them.

3.1.6 GLOSA Model

The GLOSA model uses SUMO's built-in Green Light Optimal Speed Advisory device, which implements the V2I mechanism described by Katsaros et al. [6]. Unlike the car-following tiers above, GLOSA is not itself a car-following model but a speed-advisory device layered on top of one. When a GLOSA-equipped vehicle approaches a signalised intersection, it receives the current signal phase and the time remaining until the next phase change; if the vehicle would arrive during a red phase, it reduces its speed so as to arrive at the moment the signal turns green, thereby avoiding an unnecessary stop.

Because GLOSA is an orthogonal mechanism, it was paired with the human-parameter IDM base ($\texttt{tau} = 1.0$, $\texttt{minGap} = 2.5$, $\texttt{speedDev} = 0.1$) so that the effect measured is attributable purely to the signal-aware speed advisory and not to a coincident change in car-following aggressiveness. The communication range is set to 300 m, following Katsaros et al. [6].

```
  <vType id="av_glosa" carFollowModel="IDM"
         tau="1.0" minGap="2.5" speedDev="0.1"
         accel="2.6" decel="2.0" length="5">
    <param key="has.glosa.device"    value="
       true"/>
    <param key="device.glosa.range" value="300
       "/>
  </vType>
```

GLOSA is only effective on networks containing traffic lights; it is therefore evaluated on the intersection and Bologna scenarios and is omitted from the ring and freeway sweeps, where its behaviour degenerates to the underlying

IDM base.

3.2 Measurement

The following metrics are collected for each simulation run. All values are computed from SUMO's built-in output values and then parsed by the Python scripts.

3.2.1 Average Speed

Average speed is recorded from SUMO's summary output, which reports the mean speed of all active vehicles at each simulation step. For each run, the reported step-level mean speeds are averaged over all steps where at least one vehicle is present in the network. The resulting value is reported in metres per second (m/s).

3.2.2 Vehicle Throughput

For the open-network scenarios, vehicle throughput is measured from SUMO's edge-based mean-data output. SUMO reports the flow on each edge over fixed aggregation intervals of 60 s. For each run, the throughput value is computed as the mean of all reported edge-flow values across all measured edges and aggregation intervals, and is reported in vehicles per hour (veh/h). For the ring-road scenario, route completion is not meaningful because vehicles remain in the closed loop for the full simulation.

3.2.3 Waiting Time

Waiting time is measured from SUMO's `tripinfo` output. For each completed trip, SUMO reports the total time that the vehicle spent below the waiting-speed threshold during its route. The per-run waiting time is computed as the mean of this value over all completed trips. The resulting value is reported in seconds per vehicle (s/veh).

3.2.4 Experimental Procedure

For each combination of AV model and penetration rate, the simulation is run three times with different random seeds. The random seed determines the assignment of vehicles as AV or human at departure, as well as SUMO's internal

stochastic processes such as the speed deviation of human drivers. The three repetitions yield mean values and standard deviations for each metric, which are reported in the results. The penetration rates tested range from 0% to 100% in 5% increments.

Four road networks are used in the experiments of this thesis: a closed single-lane ring, an open freeway based on the I-80 freeway, an intersection network containing multiple signalised and unsignalised intersections and a large-scale urban network of the Italian city of Bologna. The ring, intersection and freeway are minimal reference scenarios that isolate a single traffic-flow phenomenon each (wave formation, intersection management and free-flow highway behaviour, respectively), while the Bologna network is the large-scale realistic case that exposes the AV tiers to the full complexity of everyday urban driving.

3.2.5 Simulation Pipeline and Reproducibility

A Python-based simulation harness was developed to execute the full experimental pipeline in a consistent and reproducible manner. The harness prepares the SUMO route files for each experiment, assigns vehicle types according to the selected AV model, penetration rate, and random seed, runs the SUMO simulations, and stores the raw output files. This ensures that all model and scenario combinations are generated using the same procedure.

For each run, the original demand is copied into a temporary route file. Vehicles are then reassigned from the human vehicle type to the AV vehicle type under test according to the specified penetration rate. The random assignment is seeded, which makes the selected set of AVs reproducible and allows different AV models to be compared using the same underlying vehicle selection. The ring-road scenario is handled by the same harness, but is executed through TraCI so that the controlled speed perturbation can be introduced after the warm-up period.

After the simulation sweep, separate Python scripts parse the SUMO output files and generate the figures used in the results chapter. The parser extracts trip-level quantities from `tripinfo` output, time-step quantities from SUMO summary output, edge-based flow measurements, and induction-loop detector data for the ring-road scenario. The processed data are written to a tabular file and then used by the plotting script to generate the penetration-rate plots and the ring-road time-space heatmaps.

The simulation harness is used only for experiment execution and data processing. The AV behaviour itself is defined by SUMO vehicle type definitions

and SUMO's built-in car-following and advisory models. The only exception is the ring-road perturbation, where TraCI is used to temporarily reduce the maximum speed of one vehicle in order to introduce a controlled disturbance. The full implementation, including the SUMO scenario files, vehicle type definitions, simulation runner, result parser, and plotting scripts, is available in the project repository: `https://github.com/ErikFrankling/bachelor`.

3.3 Scenarios

3.3.1 Ring

The ring-road scenario is based on the ACC/CACC evaluation setup used by Porfyri, Mintsis, and Mitsakis [12]. The network is a closed single-lane circular road with a total length of $4\ \mathrm{km}$. In the implemented SUMO network, the ring is divided into 80 edges of approximately $50\ \mathrm{m}$ each. The demand consists of 200 vehicles inserted at the start of the simulation, giving an average density of $50\ \mathrm{veh/km}$. Since the network is closed, vehicles remain on the ring and repeatedly traverse the same route. Figure 3.2 shows the resulting network as rendered by `sumo-gui`.

The simulation is run for $4000\ \mathrm{s}$ with a step length of $0.1\ \mathrm{s}$. The first $2000\ \mathrm{s}$ are treated as a warm-up period. After this warm-up, one vehicle is perturbed for $60\ \mathrm{s}$ by temporarily reducing its maximum speed. This follows the ring-road disturbance procedure used by Porfyri, Mintsis, and Mitsakis [12], where a local disturbance is introduced after traffic has reached an approximately steady state. The purpose of the scenario is to test whether different vehicle models dampen or amplify the resulting speed disturbance.

Because the ring road is a closed network, trip completion metrics are not meaningful in the same way as for open networks. The scenario therefore includes 80 induction-loop detectors placed at $50\ \mathrm{m}$ intervals around the ring, matching the segmentation of the network. Detector data are aggregated every $50\ \mathrm{s}$ and used as the primary measurement source for the ring-road analysis. This provides local speed and flow measurements around the full loop, which are used for the speed heatmap graph in results that visualizes disturbance propagation and string stability.

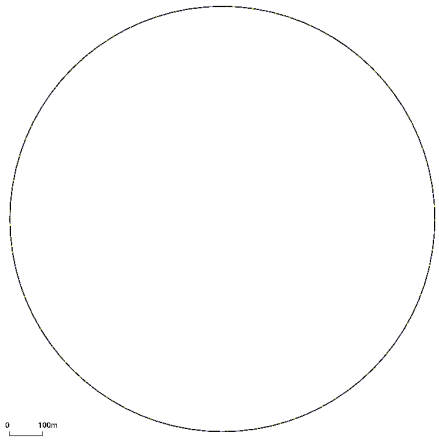


Figure 3.2: The ring scenario, a closed single-lane loop.

3.3.2 I-80 freeway

The I-80 scenario uses the open SUMO model of Interstate 80 Eastbound released by Gao, Cheng, and Zhang [17] as part of their variable speed limit study. The model covers a 10 mile section of I-80 passing through Davis, California, with demand reconstructed from PeMS loop detector data for the afternoon peak. The corridor includes a mainline source, several on-ramps and off-ramps, and a lane drop from six to three lanes roughly midway through the network, which produces recurrent congestion at multiple bottlenecks during the simulated period.

The network and demand structure are preserved from the original release. Heavy vehicles are removed from the demand and the remaining passenger cars are reassigned to the human vehicle type used throughout this thesis. As in the ring and Bologna scenarios, the only quantity that varies between runs is the fraction of those cars reassigned to the AV type under test.

The scenario serves as the realistic freeway test case in this thesis. It complements the ring road, which isolates string-stability behaviour on a closed single-lane loop, by adding multi-lane traffic, geometric bottlenecks and a realistic demand profile.

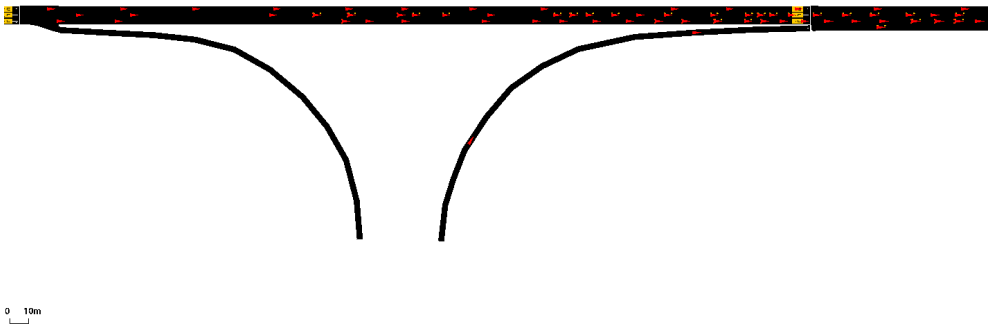


Figure 3.3: Section of the I-80 Eastbound network during a simulation run. Vehicles are rendered as red arrows.

3.3.3 Bologna

The Bologna scenario is the real-world urban network released by Bieker et al. [18] as an open, validated SUMO test case modelling a section of the Italian city of Bologna. It contains signalised intersections, roundabouts, multi-lane arterials, turning movements, and heterogeneous demand, and is used in this thesis as the main large-scale benchmark where the AV tiers are exposed to the full combination of effects that characterise everyday city traffic. The network, traffic signals, and demand are taken directly from Bieker et al. [18] without any structural modification; only the vehicle types attached to the demand are replaced according to the penetration rate of the current run. Figure 3.4 shows the overall street layout of the network and Figure 3.5 shows a close-up of a representative signalised intersection during a run.

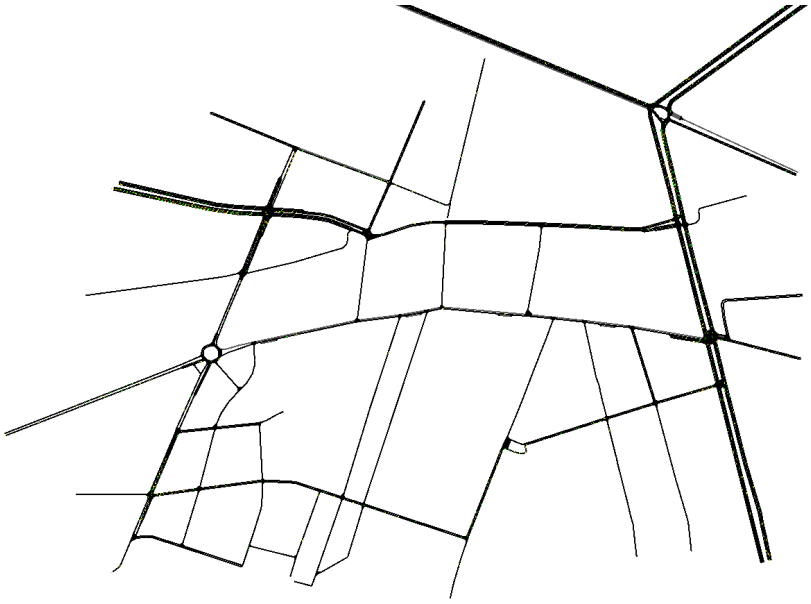


Figure 3.4: Overview of the Bologna urban network used as the main large-scale benchmark in this thesis.

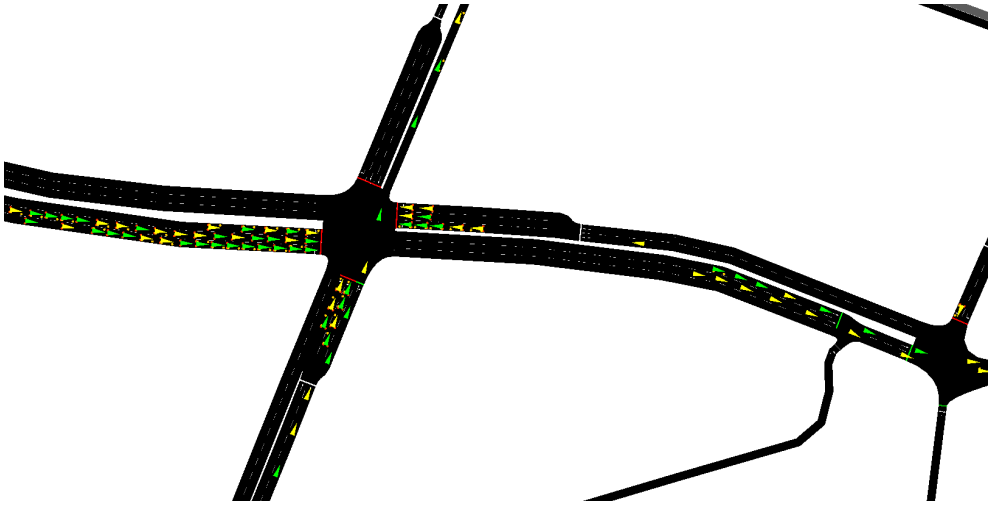


Figure 3.5: Close-up of a signalised intersection in the Bologna network during a simulation run.

3.3.4 Intersection

The intersection scenario is a 4x3 road grid network created specifically for this thesis to evaluate how each AV model performs in an environment dom-

inated by intersections. The intersection scenario is designed to expose the models to the kind of decision-making complexity that arises at junctions. The network consists of a regular grid containing a mix of unsignalised priority intersections and signalised intersections with traffic lights. Vehicles enter the network at the boundary edges and are assigned random routes through the grid, producing a varied demand pattern across the network.

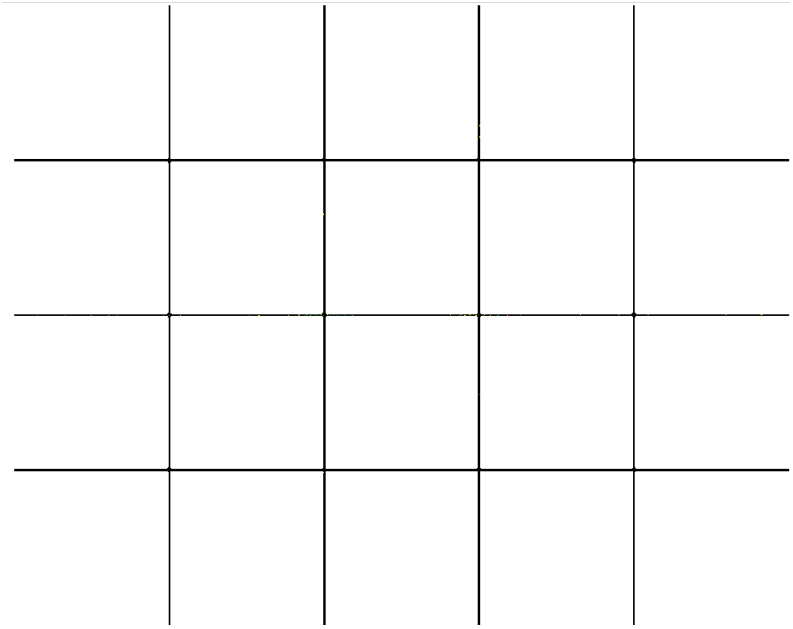


Figure 3.6: Overview of the Intersection scenario network.

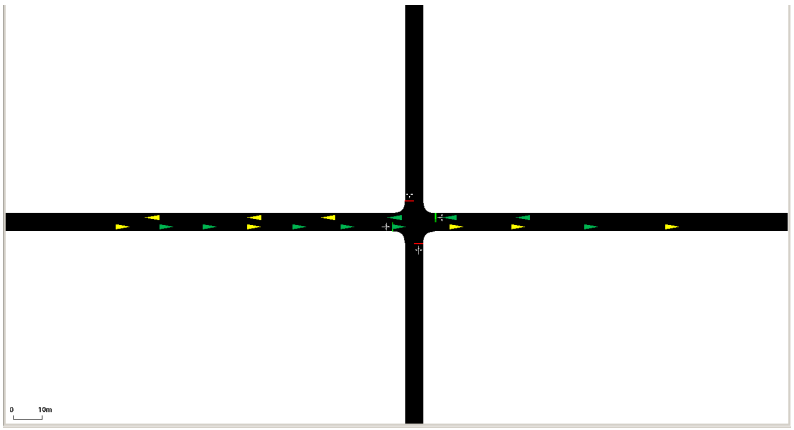


Figure 3.7: Close-up of a signalised intersection in the Intersection network during a simulation run with 50% penetration of GLOSA AVs.

Chapter 4

Results

This chapter presents the outcome of the penetration-rate sweep across the four scenarios introduced in the Method chapter. Every curve shows the mean across the three random seeds as a function of penetration rate and the horizontal dotted line marks the 0% penetration (human baseline) value. The metrics shown are the ones defined in the Background and Method chapters, with a single waiting time plot included for the intersection scenario.

4.1 Ring Road

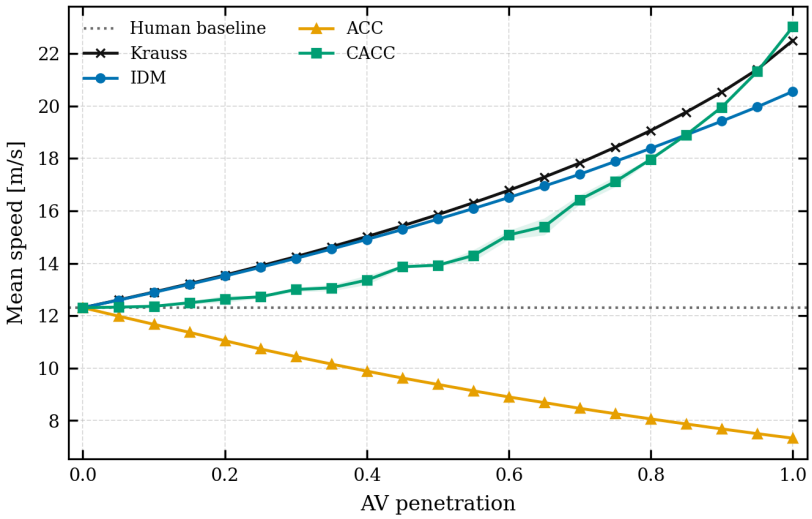


Figure 4.1: Mean speed on the ring scenario as a function of AV penetration rate

Figure 4.1 shows how the mean network speed on the ring evolves as the AV share is increased from 0% to 100%. The vertical axis is the average vehicle speed reported by SUMO's summary output, averaged over the full simulation. The Krauß, IDM and CACC tiers perform similiarly and increase the mean speed with as much as 85% for CACC while ACC results in a decrease of almost 40% in mean speed for increasing penetration rates.

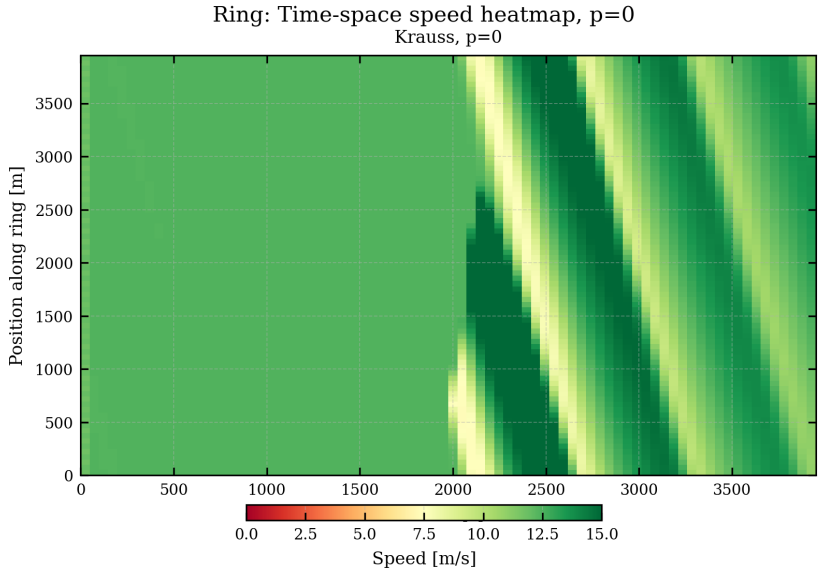


Figure 4.2: Time-space speed heatmap on the ring at 0% penetration (the shared human baseline). The vertical axis is position along the loop and the horizontal axis is simulation time; colour encodes local speed, with green indicating free-flow speeds and yellow to red indicating the slow-moving regions that make up a stop-and-go wave. The disturbance is introduced at the end of the 2000 s warm-up and propagates backwards around the loop as a persistent wave. Since all three tiers are identical at 0% penetration, only one baseline panel is shown.

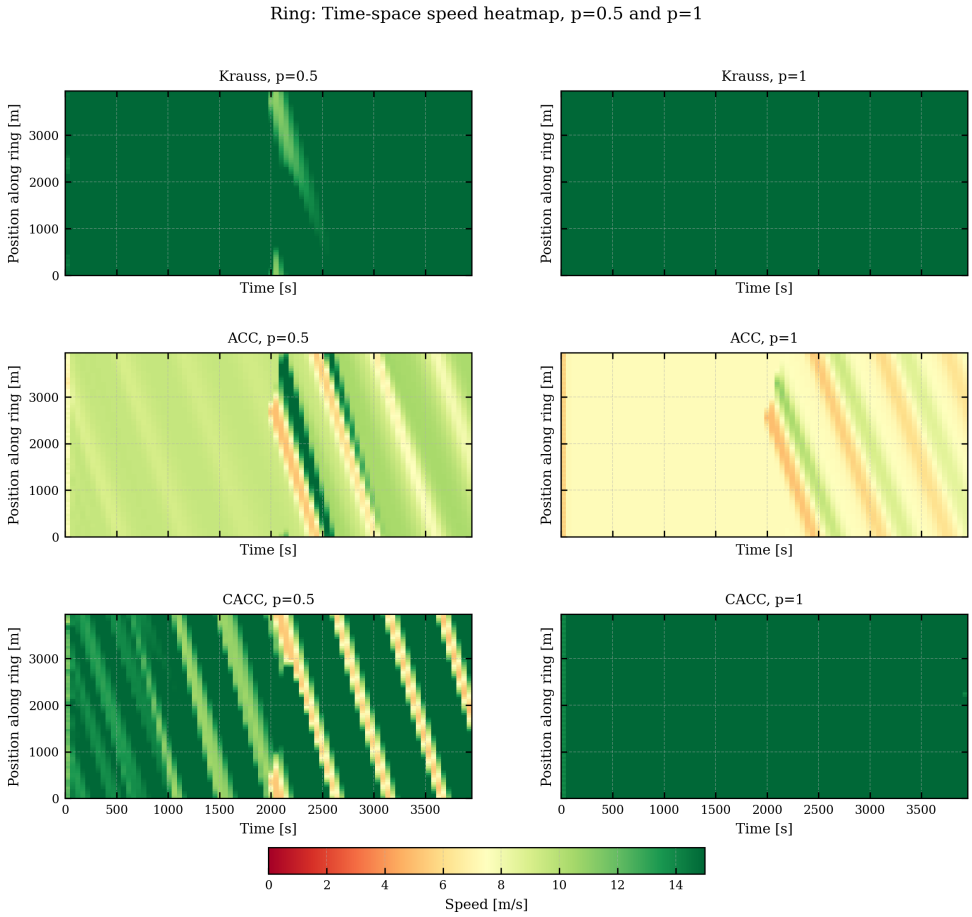


Figure 4.3: Time-space speed heatmaps on the ring for the Krauß, ACC and CACC tiers at 50% ($p = 0.5$) and 100% ($p = 1$) penetration. Axes and colour scale match Figure 4.2. The same disturbance as in the baseline is introduced at the end of the warm-up.

Figures 4.2 and 4.3 show the time-space speed field on the ring. Figure 4.2 is the shared human baseline at 0% penetration, identical for all three tiers: the deliberate perturbation introduced at the end of the warm-up appears as a slow-moving region that propagates backwards around the loop, producing the diagonal stop-and-go bands that persist for the remainder of the run. This is the phantom traffic jam behaviour described in the Background chapter, reproduced here as a controlled disturbance.

Figure 4.3 shows how each tier responds as penetration increases. The two deterministic tiers suppress the disturbance: at 100% penetration both the Krauß and CACC fields are almost uniformly green, indicating that the

injected wave is fully dissipated and the loop returns to a smooth high-speed state. At 50% penetration the Krauß tier already removes most of the banding, leaving only a single faint residual wave, whereas the CACC tier still shows pronounced diagonal bands; the mixed stream of human and CACC vehicles, in which CACC-to-CACC pairs are not yet dominant, dampens the disturbance less effectively until full penetration is reached.

The ACC tier behaves differently. Rather than dissipating the wave, it lowers the overall speed of the loop: the field shifts from green towards yellow as penetration increases, and faint diagonal bands remain visible even at 100%. This is consistent with the decreasing mean speed reported for ACC in Figure 4.1, and reflects the bounded acceleration and longer time gap of the empirically calibrated ACC controller rather than an ability to stabilise the traffic stream.

4.2 I-80 freeway

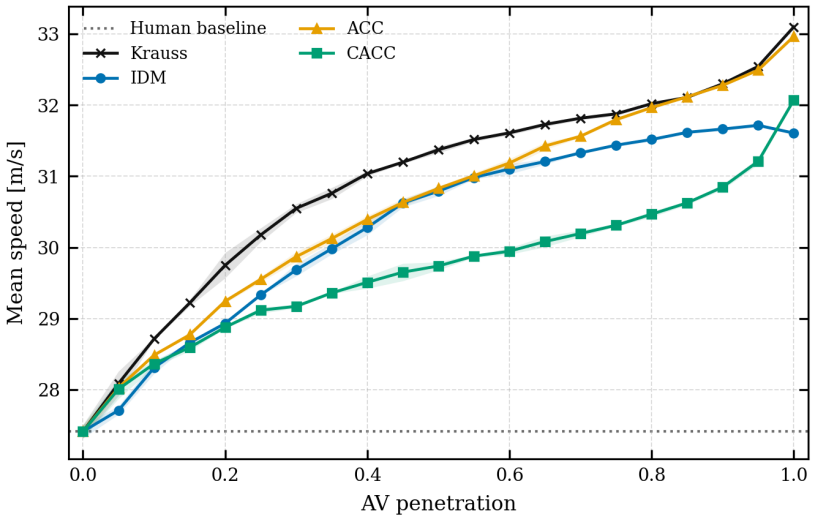


Figure 4.4: Mean speed on the I-80 freeway scenario as a function of AV penetration rate.

Figure 4.4 reports the mean speed on the freeway scenario. The mean speed is increased for all types of AVs for increased penetration rates. The best performing model, Krauß, increases mean speed by 20% while the worst performing model, IDM, reports a 15% increase. All the models except IDM shows a significant increase when approaching 100% penetration.

4.3 Bologna

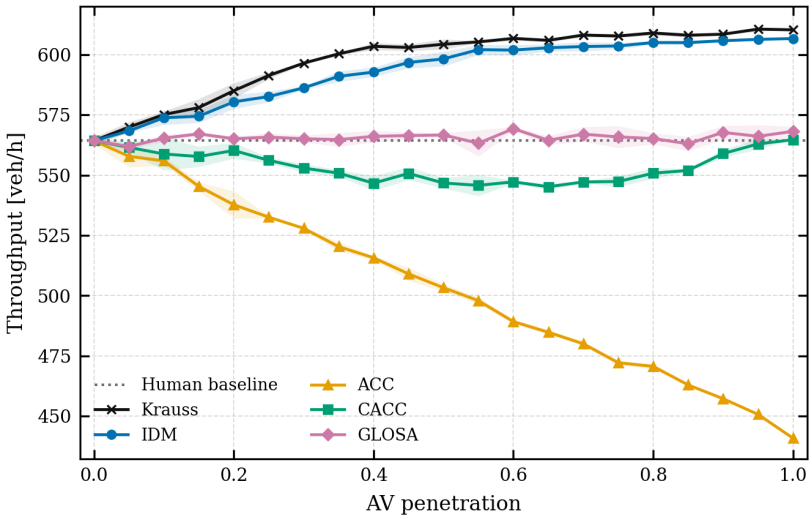


Figure 4.5: Throughput on the Bologna scenario as a function of AV penetration rate.

Figure 4.5 shows the throughput metric on Bologna, measured from SUMO's edge-based mean-data output as described in the Method chapter, and corresponds to the throughput metric Q defined in the Background chapter. While the ACC model showed regressing throughput, the IDM, and Krauß models demonstrated higher throughput with both reaching almost 7.5% increase in throughput. GLOSA performs at the human baseline while CACC initially performs worse to return back to the human baseline at 100% penetration.

4.4 Intersection

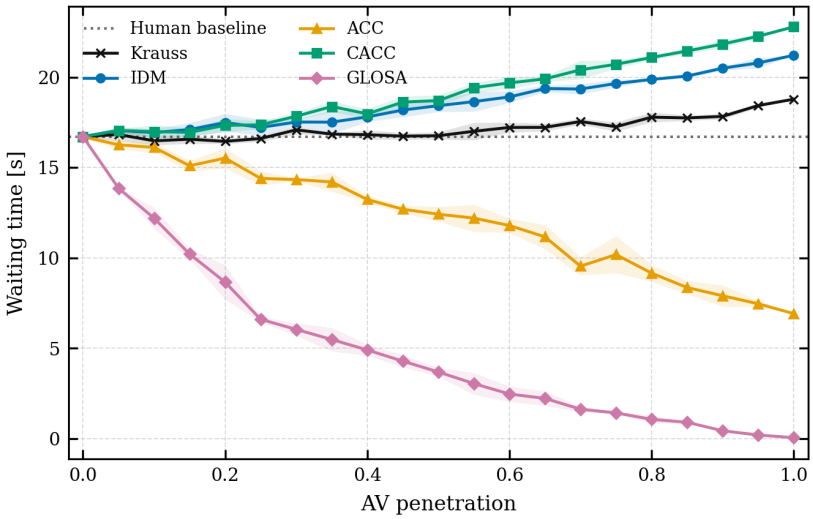


Figure 4.6: Mean waiting time as a function of penetration rate.

Figure 4.6 reports the mean waiting time per trip on the intersection grid, extracted from SUMO's tripinfo output as the time each vehicle spent stationary. Since the trips are the same across runs it shows which models effectively plan their speed to not stop before intersections. The GLOSA model shows up to a 100% decrease since it times the arrival to each intersection and therefore eliminates waiting time. ACC show a 60% decrease while Krauß, CACC and IDM each increasing waiting time and with CACC reporting up to a 45% increase.

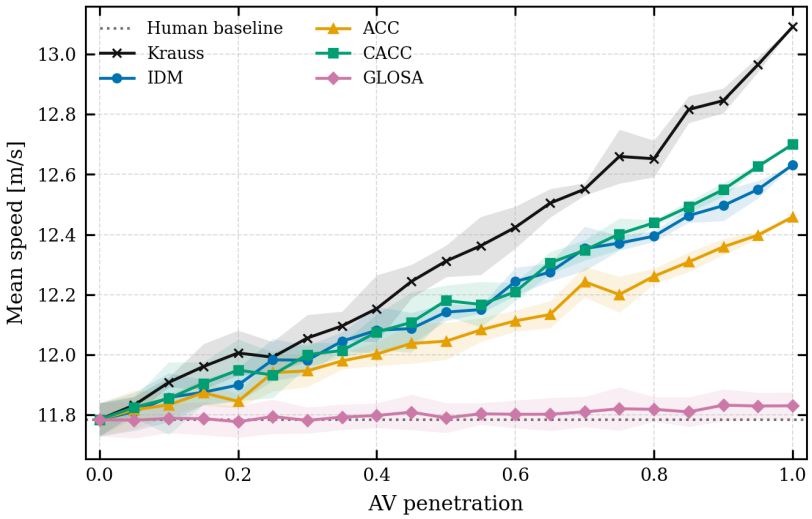


Figure 4.7: Mean speed as a function of penetration rate.

Figure 4.7 shows the mean speed on the intersection grid. The GLOSA model keeps slightly above the human baseline while all the other models show a trend of increasing speed with Krauß achieving the highest speed with a 10% increase.

Chapter 5

Discussion

This chapter discusses the outcomes of the penetration-rate sweep presented in Chapter 4 and addresses the research questions posed in the introduction. The resulting numbers should be approached at a high level rather than as direct predictions of real-world deployment, since real driving involves heterogeneity in vehicle types, driver behaviour, and conditions that are not captured by any of the car-following models examined here.

The five AV tiers studied in this thesis are also not strictly numerically comparable to one another, as established in the Method chapter. The IDM-based tiers and the Krauß tier rest on analytical car-following formulations, the ACC and CACC tiers on empirically calibrated controllers from production hardware, and GLOSA is a speed-advisory mechanism layered on top of an existing car-following model. They also do not share a common longitudinal envelope: the ACC and CACC tiers follow the bounded acceleration and maximum-speed values of Porfyri, Mintsis, and Mitsakis [12], while the IDM-based tiers use the shared envelope defined for them. The tiers appear together on the same plots, but the value of doing so lies in showing how the broader picture of AV integration takes shape depending on which aspect one chooses to model, not in comparing the tiers as direct numerical alternatives. The findings below are framed accordingly.

Sections 5.1-5.3 address the three sub-questions in turn, Section 5.4 synthesises them into a combined answer to the core research question, and Section 5.5 covers the limitations of the study.

5.1 RQ1: Comparison of AV Control Models

The first research question is formulated as follows:

How do different AV control models (parameter-tuned car-following, ACC, CACC, GLOSA) compare in their effect on traffic flow?

The choice of car-following model has a clear effect on the measured outcome, and the ranking of models is not consistent across networks. On Bologna, the deterministic Krauß and IDM tiers produced the largest improvements in mean speed and throughput. CACC sat only slightly above the baseline, while GLOSA matched it. The GLOSA tier uses the same IDM car-following model and the same parameters as the human baseline, with the speed advisory device added on top. The comparison against the human baseline therefore measures the benefit of GLOSA over normal human driving directly. On Bologna this benefit is small. Signal timing is not the dominant constraint on throughput on this network, and the expected benefits of GLOSA are mainly reduced energy consumption rather than in increased system-wide flow.

The more meaningful comparison for CACC is against ACC, since the two share the same underlying empirical car-following equations of Milanés and Shladover [9] and differ only in the V2V cooperative component. Against that reference, CACC produced a substantial improvement, in line with the findings of Porfyri, Mintsis, and Mitsakis [12], whereas the gap to the IDM-based baseline is small and partly reflects the fact that the two are governed by entirely different car-following equations. ACC, being a driver-assist feature rather than a full AV system, sat below the baseline at non-zero penetration, which matches its lack of the super-human properties that distinguish the other tiers from a human driver.

The Krauß and IDM tiers produced the largest improvements among the models tested. The Krauß tier uses the same parameter set as Lu et al. [5], and its behaviour on Bologna is in line with the capacity increases reported in that study.

5.2 RQ2: Effect of Penetration Rate

The second research question is formulated as follows:

How do different levels of AV penetration (0%, 5%, 10%, . . . , 100%) affect traffic flow?

The hypothesis stated in the introduction was that increasing AV penetration would generally result in improved traffic flow, with the size of the effect depending on the control model and the largest marginal improvements expected between 50% and 80%. For the Krauß, IDM models on the Bologna scenario, the trend is broadly monotonic and the curves flatten at high penetration, which supports the first part of the hypothesis. The trend of the GLOSA

and CACC models is however more difficult to determine since the standard deviation being too high results in uncertain results which trends slightly better than the human baseline.

The shape of the improvement is however not as predicted for the ring road scenario which instead generally shows an increasing improvement for higher penetration rates. This might be attributed to the simplicity of that scenario leading to larger impact of the unpredictability of the human drivers which when replaced result in an increasing impact on traffic flow. This is consistent with the time-space heatmaps in Figures 4.2 and 4.3, where each added deterministic vehicle removes a further source of the disturbance, so the wave is dissipated more completely as penetration rises rather than at a single threshold.

5.3 RQ3: Influence of Road Network Type

The third research question is formulated as follows:

How does the road network type (closed loop, free-flow highway, signalised urban) influence the effect of AV adoption on traffic flow?

The ring road and the freeway expose the models to broadly similar free-flow conditions, with most tiers improving mean speed at increasing penetration rates. The exception is ACC on the ring, whose mean speed falls below the baseline (Figure 4.1); on the freeway all tiers improve. The key structural difference is that the I-80 freeway is a multi-lane corridor with on- and off-ramps and a lane drop from six to three lanes, introducing geometric bottlenecks and lane-changing behaviour absent from the single-lane ring. Despite this added complexity, the relative ranking of models remains similar across both scenarios. The dominant mechanism on both networks appears to be the same, that replacing unpredictable human drivers reduces wave formation. The additional geometry of the freeway does not substantially alter how the models compare.

The intersection also shows an increase for all models with one of the differences being it includes the GLOSA model which performs right above the baseline. However the GLOSA eliminates waiting time at 100% penetration showing that the speed advisory system works perfectly and results in no stops at intersections. While this does not reflect on the overall performance, it could lead to less energy used as less accelerating/decelerating is required. The other significant difference being that the Krauß model performs at higher rate than the other models showing that it can handle these types of scenarios very well.

Bologna is where the differences between models actually appear. The

combination of signalised intersections, turning movements and heterogeneous demand exposes each tier to a wider range of conditions, and the resulting curves are the most informative in the thesis. The deterministic tiers improve throughput substantially, while CACC and GLOSA stay near the baseline. This is not visible on the simpler networks, which suggests that single-phenomenon benchmarks such as a ring or a straight freeway are not on their own sufficient to characterise the effect of an AV control model.

The time-space heatmaps in Figures 4.2 and 4.3 make the mechanism behind these ring-road results explicit. The Krauß and CACC tiers visibly absorb the injected disturbance, with the stop-and-go bands present in the baseline disappearing entirely by 100% penetration, which directly explains the large mean-speed gains of those tiers on this network. The ACC tier instead lowers the loop speed without removing the bands, which is why it is the only tier whose mean speed falls below the baseline. The heatmaps therefore confirm that the ring improvements come specifically from wave dissipation rather than from higher cruising speeds, and show that this dissipation is only partial in mixed traffic and reaches its full effect at high penetration.

5.4 Core Research Question

Returning to the core research question, microscopic traffic simulation reveals that AVs have the potential to improve traffic flow, but the answer depends on three correlated choices: how an AV is modelled, how widely it is deployed, and the network it operates on.

The model determines which aspect of automation is tested. Removing human reaction noise (IDM, Krauß), adding an empirically calibrated cruise controller with or without V2V coupling (ACC, CACC), and adding signal-coordinated driving (GLOSA) are five different things, and the simulation shows that each produces a different shape of effect. The penetration rate determines how much of the traffic stream the chosen model can act on, but the shape of the resulting curve still depends on the network: bottleneck-dominated networks benefit most near full penetration, the urban network saturates earlier, and the intersection scenario shows GLOSA eliminating waiting time at signals without producing a comparable effect on overall traffic flow. The network in turn determines which aspect of automation is exercised at all, so the same tier can produce a large effect on one network and none on another.

The hypothesis that increased AV penetration improves traffic flow is supported in the broad sense, but qualified in all of the ways above. The contribution of this thesis is not a single number for the benefit of AV adoption but

a map of how that benefit takes shape across modelling choices, deployment scales and network types.

5.5 Limitations

The AV models used in this study are simplified abstractions and should not be interpreted as accurate representations of either human drivers or real autonomous vehicles. Each tier captures a narrow set of properties that an AV could in principle exhibit, such as reduced reaction noise, shorter time headways, V2V cooperation, or signal-aware speed control, and isolates these properties under controlled simulation conditions. The results are intended to indicate the direction and rough magnitude of the effects that increasing AV penetration may produce, but it is not established that the chosen abstractions are appropriate, and the observed trends may not correspond to those of a real-world deployment.

A further consideration is that the tiers compared in this study differ substantially in their mathematical formulation. The Krauß and IDM models are derived from analytical principles, the ACC and CACC models are calibrated against field measurements of production hardware, and GLOSA is a speed-advisory device layered on top of an existing car-following model. Differences in the measured outcomes therefore reflect both the AV-specific properties of each tier and the underlying differences between the car-following formulations themselves. The results should be interpreted with this in mind, and direct numerical comparisons between tiers should be read as comparisons between modeling approaches rather than between AV technologies as such.

The AV models are simplified representations of fully autonomous vehicles, and each tier is evaluated independently with no simulation containing a mixture of different AV types. A real-world rollout would include vehicles at different stages of capability, and the interaction between those technologies is not assessed here. Safety functionality is likewise assumed to be satisfied.

Only passenger vehicles are considered. Heavy goods vehicles, buses and non-motorised road users are absent from all four networks, which is a particularly strong simplification for Bologna and I-80.

The study was also limited by available hardware, as each configuration was evaluated over three random seeds. A larger number of seeds would yield more statistically correct results, though the trends observed would likely remain consistent. It serves as an indication on the performance of each model across different scenarios but in direct comparisons and for exact improvements the results should be considered with this in mind.

Chapter 6

Conclusions

This thesis used microscopic traffic simulation in SUMO to evaluate how increasing AV penetration affects traffic flow across four distinct road networks and five AV control models. The penetration sweep ranged from 0% to 100% on a closed ring, an open multi-lane freeway, the realistic urban network of Bologna and the intersection grid. The AV control models were the Krauß, IDM, GLOSA, and the ACC and CACC cruise controllers.

As discussed, the choice of AV control model has a clear effect on the measured outcome, and the ranking of models is not consistent across networks. On Bologna, the deterministic IDM and Krauß tiers produced the largest improvements, while CACC and GLOSA tracked close to the human baseline. CACC outperformed ACC on Bologna, in line with the expectation that the V2V component contributes beyond the standalone ACC behaviour. The simpler ring and I-80 freeway scenarios were unable to differentiate between the models to the same extent as the urban network while the intersection model differentiated Krauß as the overall best performing model.

The ring-road time-space heatmaps further showed that the Krauß and CACC tiers fully dissipate an injected stop-and-go wave at high penetration, while ACC reduces speed without removing the wave, confirming that the improvements observed on the closed loop stem from wave dissipation rather than from higher cruising speeds.

The hypothesis that increased AV penetration would improve traffic flow is therefore supported by the results. It holds for all the AV control models, and the magnitude of the effect depends strongly on both the model used and the network on which it is evaluated. That the largest improvements were seen between 50% and 80% penetration rate is not supported by this study. Instead, the results continue to improve with the largest improvement generally seen

when reaching 100%.

6.1 Societal Impact

The findings of this thesis have implications beyond the immediate scope of traffic simulation. As autonomous vehicles continue to be deployed in real-world environments, understanding which control models and at what penetration rates they begin to meaningfully improve traffic flow is relevant to urban planners, policymakers, and transport authorities making decisions about infrastructure investment and AV regulation.

This study provides actionable insights into AV deployment planning by highlighting that meaningful benefits require high penetration rates and depend strongly on both model choice and network type. Accounting for these factors when designing rollout strategies could help cities and transport authorities better manage the gradual transition toward autonomous traffic, ultimately contributing to more efficient and less congested urban environments for all road users.

6.2 Future Work

The most direct extension is to evaluate mixed AV deployments in which different control models coexist in the same simulation, since the models in this study were each evaluated in isolation. A real-world rollout would include vehicles operating at different levels of automation simultaneously, and the interaction between those technologies is not assessed here.

Vehicle type heterogeneity is another natural extension. Introducing heavy goods vehicles, buses and non-motorised road users would allow a more realistic assessment of the marginal effect of AVs in mixed traffic. Finally, reinforcement-learning controllers of the kind developed by Wu et al. [10] were out of scope for this thesis and would be a useful counterpoint to the models studied here.

Another interesting aspect to study would be CO2 emissions or energy used as it provides another aspect of what a well-performing model could accomplish. For example the GLOSA model eliminated waiting making less acceleration/deceleration necessary and eliminating idling.

A further extension is to evaluate more sophisticated AV control models. The five models studied here are simplistic, while real-world AV systems are likely to be considerably more complex, for example in how they make use of

V2V and V2I communication. Future work in this direction would give more useful insight into the effects that AV adoption could have on traffic flow.

6.3 Sustainability considerations

The primary sustainability relevance is the relation between smoother traffic flow and fuel consumption. Traffic waves result in braking and accelerating which are known to increase fuel consumption, so suppressing them leads to more efficient transportation. The ring scenario, shown in Figures 4.2 and 4.3, demonstrates that the Krauß and CACC models dissipate the injected wave at high penetration, creating a more stable traffic flow and thus reducing fuel consumption, whereas the ACC model does not remove the wave.

Vehicle idling is a well-documented contributor to elevated fuel consumption per kilometre. This effect is compounded by the fuel cost of the stop itself since a full stop could have been avoided by instead reducing speed gradually in anticipation of the stop, thereby eliminating the braking event and the idle period entirely. The GLOSA model mitigates this by keeping vehicles at a reduced speed through the approach, replacing the stopping and idling with uninterrupted low-speed movement. The remaining AV models, by contrast, increase intersection waiting time and with it fuel consumption. A deployment combining GLOSA with one of these models would therefore capture the benefits of both smoother signal approach and better traffic wave suppression.

Reduced fuel consumption directly results in less emissions, as a regular combustion engine produces CO2 and other exhaust emissions. Reducing these emissions would contribute to the 13th UN sustainability goal of climate action by reducing the CO2 produced combined with contributing to the 11th UN sustainability goal of sustainable cities and communities by reducing emissions of harmful pollutants such as nitrogen oxides and particulate matter, which affect air quality in dense urban environments where traffic congestion is common.

6.4 Ethical implications

The largest ethical implication concerning AVs is related to their safety as a human driver can be held accountable for their decisions while driving though, in a driver-less vehicle the accountability of accidents and decisions is unclear. The way an AV would react would be according to pre-programmed ethics in the automation model but as the developers can not be held accountable the

ethics must therefore be decided on a legal level. Thus in case of accident the AV would be blameless. This consideration might be overlooked if the AVs improve safety by a considerable margin however, as of now most self-driving vehicles require a human driver's attention to ensure safety[19]. These implications are not addressed in this thesis as it only evaluates the traffic flow effects of AV control models. A large scale driver-less real-world implementation of the level AV models observed in this study, assuming an equivalent real-world model exists, would require further development and legislation.

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